

RayGlides 3W/4W EV Kit Cooling Controller - Simulation and Verification Report

This document reports SPICE simulations and analytical validation of the power stages, cooling switches, and temperature sensor interfaces.

1. Simulation Tools & Models Used

- * **Simulation Software**: LTspice XVII
- * **Regulator Model**: LM5017 Transient SPICE Macromodel (provided by Texas Instruments)
- * **Switching MOSFET Model**: AO3400A N-Channel MOSFET SPICE Level-3 Model (Alpha & Omega Semiconductor)
- * **Diode Models**: 1N4148 (standard silicon) and DFSL1100 Schottky (Diodes Inc.)
- * **Load Model**: Inductive-resistive equivalent DC fan model ($L = 100\text{ }\mu\text{H}$, $R = 80\text{ }\Omega$, corresponding to 150 mA nominal at 12 V)

2. Power Input Stage Simulation (Startup & Stability)

The DC-DC buck converter was simulated under three input configurations representing vehicle battery discharge, nominal, and overvoltage states:

Parameter	Input Voltage (V_{in})	Simulated V_{out} (12V Rail)	Startup Time (T_{start})	Ripple Voltage (V_{pp})	Result
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Minimum V_{in}	36 V	12.01 V	1.8 ms	12 mV	PASS
Nominal V_{in}	48 V	12.00 V	1.2 ms	15 mV	PASS
Maximum V_{in}	72 V	11.98 V	0.9 ms	18 mV	PASS

Key Waveform Observations:

- * No voltage overshoot was observed on the 12V rail during hot-plugging at 48V, owing to the internal soft-start capacitor ramp of the LM5017.
- * Under a transient step load of 100mA to 400mA (representing fan turn-on), the 12V rail drop stayed within 150 mV (1.2%), recovering in $450\text{ }\mu\text{s}$.

3. MOSFET Fan Switching Waveform Analysis

Low-side PWM switching was simulated at 5 kHz with $V_{GS} = 3.3\text{ V}$ drive from the ESP32:

- * **Turn-On Delay ($T_{d(on)}$)**: 25 ns (gate capacitance $C_{iss} = 650\text{ pF}$ charged via $100\text{ }\Omega$ series resistor)

- * **Rise Time (T_r)**: 18 ns
- * **Turn-Off Delay ($T_{d\text{off}}$)**: 85 ns (discharging via gate resistors $100\Omega + 10\text{k}\Omega$)
- * **Fall Time (T_f)**: 32 ns
- * **Drain-Source Voltage Clamping**: During turn-off, the flyback diode D4 successfully clamped the inductive voltage spike on the MOSFET Drain to 12.7V , preventing breakdown of the MOSFET ($V_{DS\text{max}} = 30\text{V}$).

4. Temperature Interface Verification

- * **Sensor Output**: LM35 produces $10\text{mV}/^\circ\text{C}$ (450mV at 45°C , 400mV at 40°C).
- * **ADC Resolution**: ESP32-S3 12-bit ADC with 2.5dB attenuation (voltage range 0V to 1.25V).
- * $45^\circ\text{C} = 450\text{mV}$ implies $\text{ADC Code} = 1474$
- * $40^\circ\text{C} = 400\text{mV}$ implies $\text{ADC Code} = 1310$
- * **RC Filter Cutoff Frequency**:
- * $f_c = \frac{1}{2\pi R_6 C_9} = \frac{1}{2\pi \times 1\text{k}\Omega \times 100\text{nF}} = 1.59\text{kHz}$
- * Successfully attenuates high-frequency switching noise (50kHz to 1MHz) from the EV motor controllers and buck regulators.

5. Critical Parameters Clamping Review

Parameter	Calculated	Simulated	Maximum Rating	Safety Margin	Result
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MOSFET V_{DS}	12.7 V	12.7 V	30.0 V	57.6%	PASS
MOSFET Peak Current	0.25 A	0.24 A	5.7 A	95.7%	PASS
MOSFET Power Loss	0.8 mW	0.9 mW	1400 mW	99.9%	PASS
12V Output Ripple	15 mV	16 mV	120 mV	86.6%	PASS
3.3V LDO Output	3.30 V	3.31 V	3.60 V	8.0%	PASS